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Application and Development Prospect of Digital Twin Technology in Aerospace

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Abstract: The digital twin can be seen as the link between the physical and digital worlds, and has received widespread attention both at home and abroad. By establishing an accurate digital model of the physical entity and the communication relationship between the physical entity and the model, the digital model and the physical entity can be mapped in both directions, so that the twin model can be modified continuously using the data of the physical entity. The twin model, after diagnosing, predicting and evaluating, can output the simulation results to the controller for state control of the physical entity, thus achieving consistency between the twin and the physical entity. This paper first analyzes the relationship between the basic concepts of the digital twin. It then introduces the necessity of digital twin in aerospace applications, focusing on the existing applications of the digital twin in the aerospace field. Compared with traditional modelling simulations, the digital twin has the advantages of shorting design cycle, high reliability, less frequent overhaul and low maintenance cost. Finally, the outlook for the future development of the digital twin is discussed, focusing on the current situation of digital twin applications in aerospace.

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Keywords: Digital twin, Internet of things, Cyber-Physical Systems, Aerospace, Research and Design, Operation maintenance, Production and manufacturing, Full life cycle.

1. INTRODUCTION

The digital twin is a multidisciplinary, multi-physics, multi-scale dynamic model that takes full advantage of the integration of physical models, operational data, and other information. It can complete the mapping of physical entities in the virtual space, thus reflecting the entire life cycle and process of the corresponding physical entities. At the same time, the digital twin's ability to collect real-time product data and bi-directional mapping between the virtual model and the physical entity allows the model to be continuously optimized and approximated to the physical entity.

Analysed at its roots, the concept of a twin originated with NASA's Apollo project[1]: Build two identical spacecraft, and the ones on Earth are called twin, reflecting (or mirroring) the status of the spacecraft which is on its mission. In 2003, Professor MICHAEL GRIEVES of the University of Michigan proposed the mirror space model in the course of product lifecycle Management[2]: a digital replica of a specific device or a set of specific devices that could abstractly represent a real device and be used as the basis for testing under real or simulated conditions. However, the concept did not receive much attention because the technology and understanding limited it at that time. In 2012, the U.S. Air Force Research Laboratory (AFRL) and NASA jointly proposed a digital twin for future aircraft and released the "Modelling, Simulation, Information Technology, and Processing" roadmap in the same year, bringing digital twin technology into the public view[3].

With the proliferation of the Industrial Internet of Things (IoT), digital twin technology has gained widespread importance in various fields. Developed in aerospace, the digital twin has begun to expand in automotive, construction, urban, healthcare, and other fields. However, the main applications are currently embodied in smart manufacturing [4][5], aerospace[6], and product development fields.

The modelling and simulation of systems is the basis for the establishment of the digital twin. In modelling and simulation for aerospace systems, much research has been conducted both at home and abroad. Different aircraft systems have been verified at four levels of the component model[7], weighing the speed and accuracy of the simulation. Hansen Company (USA) has developed a simulation system for aircraft electrical systems, and the integrated simulation of aircraft electrical systems is also an essential part of the multi-electric aircraft program in Europe. The U.S. Air Force Research Laboratory (AFRL) proposed the Integrated Aircraft Energy Technology (INVENT) program in 2008, which advocates the development of model-based design techniques that use models to simulate the whole aircraft system and improve the energy efficiency of the whole aircraft.

At present, many universities in China have carried out simulation modelling and verification[8] work for aerospace systems, including Nanjing University of Aeronautics and Astronautics, Beijing University of Aeronautics and Astronautics, and Northwest Polytechnical University, which are representative.

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① A digital simulation system for aircraft power supply system is developed. ②The configuration method of aircraft hybrid power supply system is researched[9]. ③ The intelligent optimization design method of power supply system structure is developed[10]. ④ The real-time semi-physical test platform of aircraft power distribution system based on RT-LAB platform was built, and the modelling and simulation technology of aircraft power system was studied, as well as the key technology of system-level digital joint simulation. ⑤ The "virtual reality + Internet" technology was integrated into the experiment of the power supply system of large multi-electric aircraft, and the virtual simulation experiment platform for power supply management and fault reconstruction of multi-electric aircraft, as shown in Figure 1, was developed for the sharing of industrial and social resources. Subsequently, a digital twin experimental verification platform for aircraft power supply system can be built based on this.

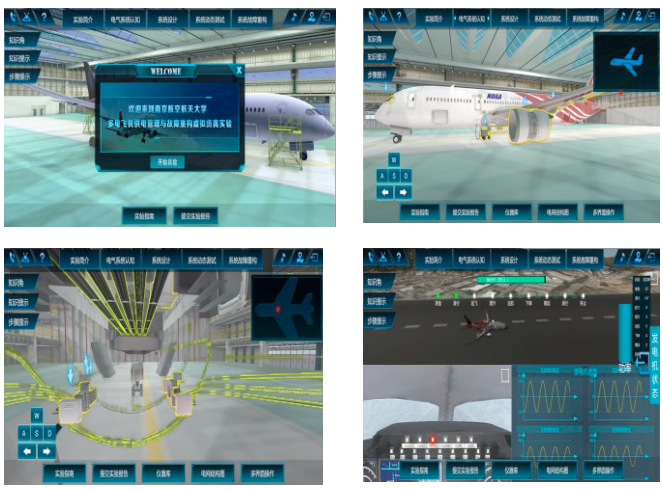

Fig. 1. Virtual simulation experiment of aircraft power supply system

The Industry 4.0 Institute classifies the digital twin into the simulation, connection and data factions based on the core of the application scenarios so that the early simulation modelling work can lay the foundation for the development of digital twin technology.

2. DIGITAL TWIN CONCEPT CLARIFICATION

In order to research digital twin technology, it is first necessary to identify the basic concepts related to digital twin technology. There are different views on the relationship between digital twin and concepts such as simulation, cyber-physical systems (CPS), and the Internet of Things (IoT). Although these concepts are closely related, they are still fundamentally different. Therefore, this section examines the relationship between this concept based on the aircraft digital twin and identifies the relationship between the digital twin and simulation, cyber-physical systems, and IoT.

2.1 Digital Twin and simulation

The digital twin is a precise mapping of physical entities in the information space and the existence of an identical physical entity. The digital twin will change in real-time according to the change of state of the physical entity, which is an inversion and mapping of the current state of the physical entity. Simulation is based on the existing entity to build a model and to simulate the possible state of the physical entity in advance, and analyse the possible state change of the physical entity based on the simulation result. Therefore, it can be seen that the digital twin is a model, a description of things, while the simulation is based on the model of various activities, there is a big difference between the simulation and digital twin, and should not be confused with each other.

2.2 Digital Twin ,CPS and IoT

The Internet of Things is based on the connection between physical assets through which data can be transmitted. Connections are made possible by the secure implementation of computer networks and communication protocols. This communication is based on typical network protocols or proprietary protocols. However, IoT does not include the concept of analytics in Cyber-Physical Systems. CPS is based on the transfer of information between physical assets, the extraction of knowledge from the information, and based on the results of the information analysis and processing to obtain control commands and re-act this control commands on the physical assets. Thus IoT can be viewed as the infrastructure for the implementation of CPS.

The digital twin can also be considered as the foundation of CPS. Establishing a digital twin model means establishing a digital virtual body based on the physical entity. The digital twin goes through a process of approximating the physical entity's shape and state until it appears to be complete harmony with the physical entity. "resemblance". This part plus the simulation of the model corresponds to the information system in the CPS, adding the simulation part and the control part of the digital virtual entity based on establishing the digital twin, i.e., based on the simulation and prediction results of the digital twin through autonomous decision making, using the controller to control the physical entity, optimizing the operation state of the physical entity and realizing the iterative optimization part of the virtual and real objects, which can constitute the completed CPS. based on the above analysis, the relationship between the aircraft digital twin and CPS and IoT can be derived, as shown in Figure 2.

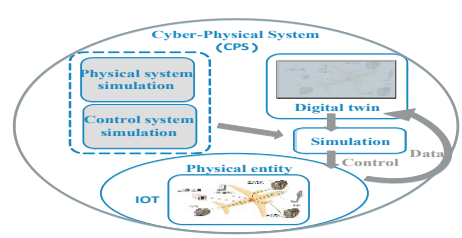

Fig. 2 The relationship of digital twin, simulation, CPS and IOT

From the above diagram, it can be seen that CPS contains three parts: digital twin, simulation and IoT. The IoT mainly includes the physical entity of the aircraft and the communication between the physical entity and the digital twin. The digital twin is constructed based on the physical entity, and the digital twin should have a high-precision simulation model corresponding to the physical entity. The digital twin can be modified based on the output data from the physical entity so that it continually approximates the physical entity and can reflect the operation state of the physical entity in real-time. The simulation part is based on the digital twin's simulation prediction. The combination of the digital twin, simulation, and IoT can form a CPS system. The digital twin and simulation correspond to the information part of CPS, while the Internet of Things (IoT) corresponds to the physical and communication parts of CPS.

Clarifying the relationship between digital twin, CPS system, Internet of Things (IoT) and simulation can help us further to understand the meaning and extension of digital twin technology and lay the foundation for further research.

3. APPLICATION OF DIGITAL TWIN IN AEROSPACE

Digital twin technology was first applied in the aerospace field. Therefore, it is important to analyse the current status of digital twin applications in aerospace and lay the foundation for applications in other fields.

3.1 Application of digital twin technology in aviation

The composition of modern aircraft is becoming more and more complex, as shown in [11]: ①the number of components that make up an aircraft is extensive, and each component has its characteristics, and the coupling between components will derive some new characteristics; ② the dynamic characteristics are strong, and the state of the components changes rapidly with time; ③the aircraft flight environment is uncertain, and the probability of complex systems suffering accidental damage in a highly uncertain environment increases, making the aircraft more vulnerable to damage. It is not easy to ensure system reliability. ④ periodic maintenance when facing a complex system, there is a lack of accurate estimation of the current state of the system, which is prone to too frequent inspection and maintenance, or premature failure of the system due to untimely maintenance, resulting in high maintenance costs and insufficient reliability of the aircraft[12]. Therefore, the emergence of the digital twin presents a new research idea to solve the above problems.

In the aviation sector, some research has been carried out on the digital twin. This research has been carried out across aircraft design and development, aircraft manufacturing, and aircraft operation and maintenance. Figure3 shows a diagram of the aircraft digital twin.

The digital twin consists of three components: the physical entity of the aircraft, the digital twin, and communication between the physical entity and the digital twin. The digital twin is continuously modified by data collected from sensors present in the physical entity, and the digital twin controls the physical entity through inversion and prediction. This allows the simulation of the entire life cycle of the aircraft from design to maintenance.

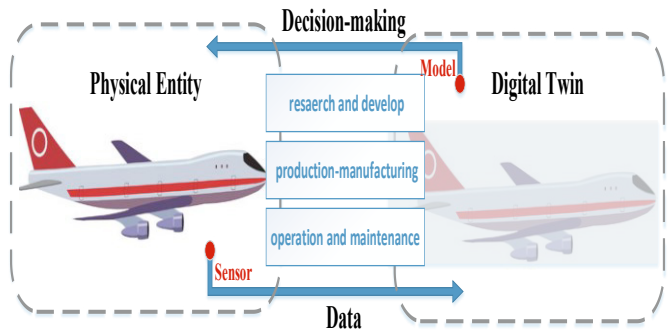

Fig. 3. A digital twin system framework for airplane

In terms of design development, the establishment of a digital twin allows for quality prediction and early identification of product defects at the early stages of design. In the digital twin, product design can be continuously optimized to reduce development costs.

In manufacturing, the establishment of the digital twin optimizes the manufacturing process, optimizes the scheduling scheme, and allows for more rational evaluation of scheduling decisions and allocation of resources.

In terms of system operation and maintenance, the development of digital twin technology can not only monitor the operation status of the product in real-time, but also predict the future working status and possible faults of the product based on the collected data, to make independent decisions, optimize the maintenance schedule, reduce the number of overhauls, and improve the reliability of product operation.

Generally speaking, the application of digital twin in aviation is still at an early stage. Developed countries such as the United States and Europe have actively applied digital twin technology in the whole life cycle of aircraft[13], and have carried out a lot of research and exploration of application practice from design[14], manufacturing[15] and operation and maintenance, and are continually seeking breakthrough technology based on this.

3.2 Application of digital twin technology in space

With the rapid development of space technology, the structure and composition of space vehicles are becoming more and more complex, and the performance and technical level are continually improving; under this circumstance, it has become an important issue to be solved in the field of space technology to ensure the more durable and stable operation of space vehicles in the complex space environment[16].

The sustained and rapid development of the defence science and technology industry and the aerospace industry over the years has led to increasing diversity, complexity and

intensification of aerospace model missions, which are characterized by short development cycles and high-quality requirements. At the same time, with the increasing number of on-orbit models and the continuous growth of spacecraft service time, the difficulty of their maintenance support is also becoming more and more prominent. Model-Based Systems Engineering (MBSE) is playing an increasingly important role in the field of spacecraft design and manufacturing[17]. It is changing people's lives. Driven by new technologies as well as the rapid development of demand, it makes functional diversity of spacecraft possible and poses a significant challenge to the spacecraft industry.

For the application of digital twin technology in the aerospace field, it can be divided into four main parts: ① overall spacecraft design, ② detailed spacecraft design, ③ spacecraft manufacturing, and ④spacecraft in-orbit control. Figure 4 shows a diagram of the spacecraft digital twin.

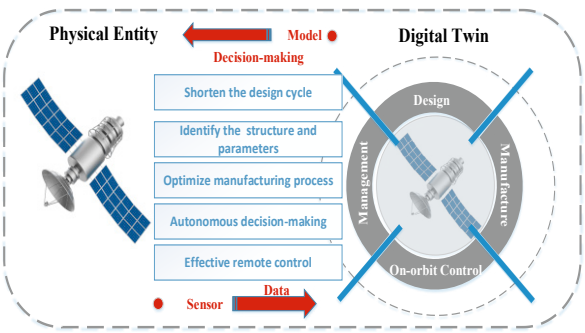

Fig.4. A digital twin system framework for space

There is a lack of understanding and communication of requirements between different departments, and it is prone to iterative problems in the design process. The application of digital twin technology in intelligent spacecraft design aims to use digital twin technology to build an information-physical fusion design simulation system[18], shorten the design cycle, and meet the demand for faster, better, and more reliable spacecraft in the overall design process.

The detailed spacecraft design is based on the conceptual requirements proposed in the overall design and theoretical knowledge of multiple disciplines to transform the conceptual design into the design of the specific structure, parameters and functions of the spacecraft, which requires the conflicting design of multiple systems and iterative iterations based on external and internal feedback and requirements. The traditional spacecraft design model is based on a document, which has problems such as inconsistent information, unclear descriptions, and communication difficulties. The establishment of a spacecraft digital twin can identify and predict the specific structure and parameters of the spacecraft at the early stage of design, and optimize the rationality of specific parameters during the design process, enabling the spacecraft design to expand more functional requirements, achieve shorter development cycle, and faster parameter iteration and optimization based on meeting predetermined targets.

Spacecraft manufacturing includes the comprehensive coordination and control of spacecraft products and the mechanical processing workshop, welding workshop, electrical assembly workshop, etc. This link determines the manufacturing quality and efficiency of the spacecraft. A great deal of research has been carried out in manufacturing, such as the study of final assembly line control architecture based on digital twin [19] and simulation and control of tooling equipment driven by digital twin[20]. However, there is still a long way to go to achieve a full element and full life cycle spacecraft digital twin. Therefore, in order to address the problems and needs faced in the spacecraft manufacturing process, the application of digital twin technology in the spacecraft manufacturing process needs to be actively carried out in order to optimize the manufacturing process, optimize the scheduling scheme, and improve the production efficiency.

In-orbit management and control of spacecraft refer to the practical application phase of their missions and services after they have entered orbit. The process includes on-orbit operations, on-orbit maintenance, on-orbit updates, fault prediction and health management. Traditional spacecraft cannot predict failures and reconstruct the system after failures occur, which can result in substantial economic losses in case of failures in actual applications. Therefore, the establishment of a spacecraft digital twin can research the prediction of spacecraft failures and health management, and based on the results of health management, autonomous decision making, autonomous management and effective remote control can be carried out to improve the reliability of spacecraft operation.

From the above analysis, we can see that digital twin technology has far-reaching implications in the aerospace industry. Throughout the whole development process and the whole life cycle of the spacecraft, digital twin technology has a profound impact on the aerospace industry. At present, the application and development of digital twin technology in the aerospace industry are still at an early stage, but it is still of great significance. Therefore, research based on the critical technology of digital twin needs to be carried out in order to meet the requirements of future more complex spacecraft.

4. THE DEVELOPMENT PROSPECT OF DIGITAL TWIN IN CHINA'S AEROSPACE FIELD

At present, China's aerospace industry is developing rapidly. Digital twin technology has significant application potential.

In aviation, a great deal of research has been conducted and some of the technologies have been applied overseas in the field of aviation digital twin technology. Among them is the Airframe Digital Twin (ADT) concept proposed by the U.S. Air Force in 2011[21], which can accurately predict the future behaviour of an aircraft and guide decision-makers to customize a management plan for each aircraft in order to extend its service life and reduce maintenance costs. The U.S. Air Force and General Electric have conducted a series of studies to implement ADT. In China, the application of digital twin in aviation is still in its infancy and has yet to involve practical applications and critical technology research.

Therefore, digital twin technology has important development potential in the aviation field.

In the aerospace field, the digital twin still has excellent potential for application. It is expected that around 2021, China will build a near-Earth space station and enter the operation and management phase[22]. The traditional spacecraft management model is to establish a corresponding ground-based physical companion system, but this approach has excellent shortcomings. First of all, a ground-based physical entity identical to the actual spacecraft needs to be built, which has high construction cost and long construction period. Moreover, the ground test cannot fully simulate the actual operating environment. Each accompanying physical system can only correspond to a fixed spacecraft, which is not universal. The above problems can be solved if digital twin technology is adopted to establish a virtual accompaniment system for the physical entities of spacecraft. The digital twin is a high-precision simulation model that corresponds to the physical entity and should have the ability to interact with the physical entity in both directions, and based on the output data of the physical entity. The digital twin can be modified so that it continually approximates the physical entity and can reflect the operation state of the physical entity in real-time. Simulation and prediction based on the digital twin can not only provide a decision reference for the operation state of the physical entity but also allow autonomous decision-making, using the controller to control the physical entity and optimize the operation state of the physical entity.

At present, there are still some challenges to developing digital twin technology in the aerospace field: ① The construction of the digital twin is a comprehensive technology involving multiple fields and levels, which itself shows the difficulty of its realization. ②There is no mature technical architecture or standard for digital twin construction in the aerospace field, so there is still a long way to go from modelling to the establishment of an intelligent and automated digital twin. ③ There is no unified specification for data management, and the lack of data may lead to errors in digital twin simulation and decision making when using data for analysis and decision making. ④In terms of virtual experiment implementation, the interaction between software and familiar interface technology is not mature, and there is no definite interaction specification between software, which makes it challenging to realize multi-domain software development, integration and operation platform.

Moreover, due to the unique nature of the aerospace industry, there are some drawbacks in conducting digital twin research, mainly in two aspects: ① Complete reliance on artificial intelligence and ignoring the importance and priority of human decision making will affect the operational safety of the aircraft; ② To bring the digital twin closer to physical entities, the number of sensors needs to be increased, thus increasing the weight of the aircraft.

5. CONCLUSIONS

This paper analyses and discusses the application of the digital twin technology in the aerospace field. Firstly, the basic concept of digital twin is introduced. After that, the current situation and application scenarios of the digital twin in the aerospace and aviation fields are explained respectively. The main points are the initial design, manufacturing and operation and maintenance of digital twin. Finally, the current research status of digital twin technology in China's aerospace field is compared with that of foreign countries, and it is concluded that the application of digital twin technology in China's aerospace field has a great prospect of development, and the challenges facing the development of digital twin technology are analysed. The theoretical foundation was laid for the subsequent research work.

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